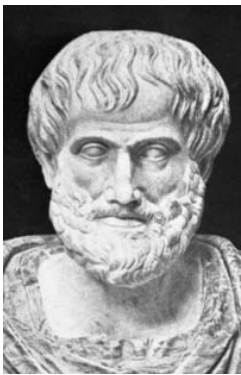


THE LOGIC OF COMPOUND STATEMENTS



Bettmann/CORBIS

Aristotle
(384 B.C.–322 B.C.)

The first great treatises on logic were written by the Greek philosopher Aristotle. They were a collection of rules for deductive reasoning that were intended to serve as a basis for the study of every branch of knowledge. In the seventeenth century, the German philosopher and mathematician Gottfried Leibniz conceived the idea of using symbols to mechanize the process of deductive reasoning in much the same way that algebraic notation had mechanized the process of reasoning about numbers and their relationships. Leibniz's idea was realized in the nineteenth century by the English mathematicians George Boole and Augustus De Morgan, who founded the modern subject of symbolic logic. With research continuing to the present day, symbolic logic has provided, among other things, the theoretical basis for many areas of computer science such as digital logic circuit design (see Sections 2.4 and 2.5), relational database theory (see Section 8.1), automata theory and computability (see Section 7.4 and Chapter 12), and artificial intelligence (see Sections 3.3, 10.1, and 10.5).

2.1 Logical Form and Logical Equivalence

Logic is a science of the necessary laws of thought, without which no employment of the understanding and the reason takes place. —Immanuel Kant, 1785

The central concept of deductive logic is the concept of argument form. An argument is a sequence of statements aimed at demonstrating the truth of an assertion. The assertion at the end of the sequence is called the *conclusion*, and the preceding statements are called *premises*. To have confidence in the conclusion that you draw from an argument, you must be sure that the premises are acceptable on their own merits or follow from other statements that are known to be true.

In logic, the form of an argument is distinguished from its content. Logical analysis won't help you determine the intrinsic merit of an argument's content, but it will help you analyze an argument's form to determine whether the truth of the conclusion follows *necessarily* from the truth of the premises. For this reason logic is sometimes defined as the science of necessary inference or the science of reasoning.

Consider the following two arguments, for example. Although their content is very different, their logical form is the same. Both arguments are *valid* in the sense that if their premises are true, then their conclusions must also be true. (In Section 2.3 you will learn how to test whether an argument is valid.)

Argument 1 If the program syntax is faulty or if program execution results in division by zero, then the computer will generate an error message. Therefore, if the computer does

not generate an error message, then the program syntax is correct and program execution does not result in division by zero.

Argument 2 If x is a real number such that $x < -2$ or $x > 2$, then $x^2 > 4$. Therefore, if $x^2 \not> 4$, then $x \not< -2$ and $x \not> 2$.

To illustrate the logical form of these arguments, we use letters of the alphabet (such as p , q , and r) to represent the component sentences and the expression “not p ” to refer to the sentence “It is not the case that p .” Then the *common logical form* of both the previous arguments is as follows:

If p or q , then r .

Therefore, if not r , then not p and not q .

Example 2.1.1 Identifying Logical Form

Fill in the blanks below so that argument (b) has the same form as argument (a). Then represent the common form of the arguments using letters to stand for component sentences.

- a. If Jane is a math major or Jane is a computer science major, then Jane will take Math 150.
Jane is a computer science major.
Therefore, Jane will take Math 150.
- b. If logic is easy or (1) , then (2) .
I will study hard.
Therefore, I will get an A in this course.

Solution

1. I (will) study hard.
2. I will get an A in this course.

Common form: If p or q , then r .

q .

Therefore, r .



Statements

Most of the definitions of formal logic have been developed so that they agree with the natural or intuitive logic used by people who have been educated to think clearly and use language carefully. The differences that exist between formal and intuitive logic are necessary to avoid ambiguity and obtain consistency.

In any mathematical theory, new terms are defined by using those that have been previously defined. However, this process has to start somewhere. A few initial terms necessarily remain undefined. In logic, the words *sentence*, *true*, and *false* are the initial undefined terms.

• Definition

A **statement** (or **proposition**) is a sentence that is true or false but not both.

For example, “Two plus two equals four” and “Two plus two equals five” are both statements, the first because it is true and the second because it is false. On the other

hand, the truth or falsity of “He is a college student” depends on the reference for the pronoun *he*. For some values of *he* the sentence is true; for others it is false. If the sentence were preceded by other sentences that made the pronoun’s reference clear, then the sentence would be a statement. Considered on its own, however, the sentence is neither true nor false, and so it is not a statement. We will discuss ways of transforming sentences of this form into statements in Section 3.1.

Similarly, “ $x + y > 0$ ” is not a statement because for some values of x and y the sentence is true, whereas for others it is false. For instance, if $x = 1$ and $y = 2$, the sentence is true; if $x = -1$ and $y = 0$, the sentence is false.

Compound Statements

We now introduce three symbols that are used to build more complicated logical expressions out of simpler ones. The symbol $\sim$ denotes *not*, $\wedge$ denotes *and*, and $\vee$ denotes *or*. Given a statement p , the sentence “ $\sim p$ ” is read “not p ” or “It is not the case that p ” and is called the **negation of p** . In some computer languages the symbol $\neg$ is used in place of $\sim$. Given another statement q , the sentence “ $p \wedge q$ ” is read “ p and q ” and is called the **conjunction of p and q** . The sentence “ $p \vee q$ ” is read “ p or q ” and is called the **disjunction of p and q** .

In expressions that include the symbol $\sim$ as well as $\wedge$ or $\vee$, the **order of operations** specifies that $\sim$ is performed first. For instance, $\sim p \wedge q = (\sim p) \wedge q$. In logical expressions, as in ordinary algebraic expressions, the order of operations can be overridden through the use of parentheses. Thus $\sim(p \wedge q)$ represents the negation of the conjunction of p and q . In this, as in most treatments of logic, the symbols $\wedge$ and $\vee$ are considered coequal in order of operation, and an expression such as $p \wedge q \vee r$ is considered ambiguous. This expression must be written as either $(p \wedge q) \vee r$ or $p \wedge (q \vee r)$ to have meaning.

A variety of English words translate into logic as $\wedge$, $\vee$, or $\sim$. For instance, the word *but* translates the same as *and* when it links two independent clauses, as in “Jim is tall but he is not heavy.” Generally, the word *but* is used in place of *and* when the part of the sentence that follows is, in some way, unexpected. Another example involves the words *neither-nor*. When Shakespeare wrote, “Neither a borrower nor a lender be,” he meant, “Do not be a borrower and do not be a lender.” So if p and q are statements, then

p but q	means	p and q
neither p nor q	means	$\sim p$ and $\sim q$.

Example 2.1.2 Translating from English to Symbols: *But* and *Neither-Nor*

Write each of the following sentences symbolically, letting h = “It is hot” and s = “It is sunny.”

- It is not hot but it is sunny.
- It is neither hot nor sunny.

Solution

- The given sentence is equivalent to “It is not hot and it is sunny,” which can be written symbolically as $\sim h \wedge s$.
- To say it is neither hot nor sunny means that it is not hot and it is not sunny. Therefore, the given sentence can be written symbolically as $\sim h \wedge \sim s$. ■

The notation for inequalities involves *and* and *or* statements. For instance, if x , a , and b are particular real numbers, then

$x \leq a$	means	$x < a$	or	$x = a$
$a \leq x \leq b$	means	$a \leq x$	and	$x \leq b$

Note that the inequality $2 \leq x \leq 1$ is not satisfied by any real numbers because

$$2 \leq x \leq 1 \quad \text{means} \quad 2 \leq x \quad \text{and} \quad x \leq 1,$$

and this is false no matter what number x happens to be. By the way, the point of specifying x , a , and b to be *particular* real numbers is to ensure that sentences such as “ $x < a$ ” and “ $x \geq b$ ” are either true or false and hence that they are statements.

Example 2.1.3 And, Or, and Inequalities

Suppose x is a particular real number. Let p , q , and r symbolize “ $0 < x$,” “ $x < 3$,” and “ $x = 3$,” respectively. Write the following inequalities symbolically:

- a. $x \leq 3$ b. $0 < x < 3$ c. $0 < x \leq 3$

Solution

- a. $q \vee r$ b. $p \wedge q$ c. $p \wedge (q \vee r)$ ■

Truth Values

In Examples 2.1.2 and 2.1.3 we built compound sentences out of component statements and the terms *not*, *and*, and *or*. If such sentences are to be statements, however, they must have well-defined **truth values**—they must be either true or false. We now define such compound sentences as statements by specifying their truth values in terms of the statements that compose them.

The negation of a statement is a statement that exactly expresses what it would mean for the statement to be false.

• Definition

If p is a statement variable, the **negation** of p is “not p ” or “It is not the case that p ” and is denoted $\sim p$. It has opposite truth value from p : if p is true, $\sim p$ is false; if p is false, $\sim p$ is true.

The truth values for negation are summarized in a *truth table*.

Truth Table for $\sim p$

p	$\sim p$
T	F
F	T

In ordinary language the sentence “It is hot and it is sunny” is understood to be true when both conditions—being hot and being sunny—are satisfied. If it is hot but not sunny, or sunny but not hot, or neither hot nor sunny, the sentence is understood to be false. The formal definition of truth values for an *and* statement agrees with this general understanding.

• Definition

If p and q are statement variables, the **conjunction** of p and q is “ p and q ,” denoted $p \wedge q$. It is true when, and only when, both p and q are true. If either p or q is false, or if both are false, $p \wedge q$ is false.

The truth values for conjunction can also be summarized in a truth table. The table is obtained by considering the four possible combinations of truth values for p and q . Each combination is displayed in one row of the table; the corresponding truth value for the whole statement is placed in the right-most column of that row. Note that the only row containing a T is the first one since the only way for an *and* statement to be true is for both component statements to be true.

Truth Table for $p \wedge q$

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

By the way, the order of truth values for p and q in the table above is TT, TF, FT, FF. It is not absolutely necessary to write the truth values in this order, although it is customary to do so. We will use this order for all truth tables involving two statement variables. In Example 2.1.5 we will show the standard order for truth tables that involve three statement variables.

In the case of disjunction—statements of the form “ p or q ”—intuitive logic offers two alternative interpretations. In ordinary language *or* is sometimes used in an exclusive sense (p or q but not both) and sometimes in an inclusive sense (p or q or both). A waiter who says you may have “coffee, tea, or milk” uses the word *or* in an exclusive sense: Extra payment is generally required if you want more than one beverage. On the other hand, a waiter who offers “cream or sugar” uses the word *or* in an inclusive sense: You are entitled to both cream and sugar if you wish to have them.

Mathematicians and logicians avoid possible ambiguity about the meaning of the word *or* by understanding it to mean the inclusive “and/or.” The symbol $\vee$ comes from the Latin word *vel*, which means *or* in its inclusive sense. To express the exclusive *or*, the phrase p or q but not both is used.

• Definition

If p and q are statement variables, the **disjunction** of p and q is “ p or q ,” denoted $p \vee q$. It is true when either p is true, or q is true, or both p and q are true; it is false only when both p and q are false.

Note The statement “ $2 \leq 2$ ” means that 2 is less than 2 or 2 equals 2. It is true because $2 = 2$.

Here is the truth table for disjunction:

Truth Table for $p \vee q$

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Evaluating the Truth of More General Compound Statements

Now that truth values have been assigned to $\sim p$, $p \wedge q$, and $p \vee q$, consider the question of assigning truth values to more complicated expressions such as $\sim p \vee q$, $(p \vee q) \wedge \sim(p \wedge q)$, and $(p \wedge q) \vee r$. Such expressions are called *statement forms* (or *propositional forms*). The close relationship between statement forms and *Boolean expressions* is discussed in Section 2.4.

• Definition

A **statement form** (or **propositional form**) is an expression made up of statement variables (such as p , q , and r) and logical connectives (such as $\sim$, $\wedge$, and $\vee$) that becomes a statement when actual statements are substituted for the component statement variables. The **truth table** for a given statement form displays the truth values that correspond to all possible combinations of truth values for its component statement variables.

To compute the truth values for a statement form, follow rules similar to those used to evaluate algebraic expressions. For each combination of truth values for the statement variables, first evaluate the expressions within the innermost parentheses, then evaluate the expressions within the next innermost set of parentheses, and so forth until you have the truth values for the complete expression.

Example 2.1.4 Truth Table for Exclusive Or

Construct the truth table for the statement form $(p \vee q) \wedge \sim(p \wedge q)$. Note that when *or* is used in its exclusive sense, the statement “ p or q ” means “ p or q but not both” or “ p or q and not both p and q ,” which translates into symbols as $(p \vee q) \wedge \sim(p \wedge q)$. This is sometimes abbreviated $p \oplus q$ or $p \text{ XOR } q$.

Solution Set up columns labeled p , q , $p \vee q$, $p \wedge q$, $\sim(p \wedge q)$, and $(p \vee q) \wedge \sim(p \wedge q)$. Fill in the p and q columns with all the logically possible combinations of T's and F's. Then use the truth tables for $\vee$ and $\wedge$ to fill in the $p \vee q$ and $p \wedge q$ columns with the appropriate truth values. Next fill in the $\sim(p \wedge q)$ column by taking the opposites of the truth values for $p \wedge q$. For example, the entry for $\sim(p \wedge q)$ in the first row is F because in the first row the truth value of $p \wedge q$ is T. Finally, fill in the $(p \vee q) \wedge \sim(p \wedge q)$ column by considering the truth table for an *and* statement together with the computed truth values for $p \vee q$ and $\sim(p \wedge q)$. For example, the entry in the first row is F because the entry for $p \vee q$ is T, the entry for $\sim(p \wedge q)$ is F, and an *and* statement is false unless both components are true. The entry in the second row is T because both components are true in this row.

Truth Table for *Exclusive Or*: $(p \vee q) \wedge \sim(p \wedge q)$

p	q	$p \vee q$	$p \wedge q$	$\sim(p \wedge q)$	$(p \vee q) \wedge \sim(p \wedge q)$
T	T	T	T	F	F
T	F	T	F	T	T
F	T	T	F	T	T
F	F	F	F	T	F

Example 2.1.5 Truth Table for $(p \wedge q) \vee \sim r$

Construct a truth table for the statement form $(p \wedge q) \vee \sim r$.

Solution Make columns headed p , q , r , $p \wedge q$, $\sim r$, and $(p \wedge q) \vee \sim r$. Enter the eight logically possible combinations of truth values for p , q , and r in the three left-most columns. Then fill in the truth values for $p \wedge q$ and for $\sim r$. Complete the table by considering the truth values for $(p \wedge q)$ and for $\sim r$ and the definition of an *or* statement. Since an *or* statement is false only when both components are false, the only rows in which the entry is F are the third, fifth, and seventh rows because those are the only rows in which the expressions $p \wedge q$ and $\sim r$ are both false. The entry for all the other rows is T.

p	q	r	$p \wedge q$	$\sim r$	$(p \wedge q) \vee \sim r$
T	T	T	T	F	T
T	T	F	T	T	T
T	F	T	F	F	F
T	F	F	F	T	T
F	T	T	F	F	F
F	T	F	F	T	T
F	F	T	F	F	F
F	F	F	F	T	T

The essential point about assigning truth values to compound statements is that it allows you—using logic alone—to judge the truth of a compound statement on the basis of your knowledge of the truth of its component parts. Logic does not help you determine the truth or falsity of the component statements. Rather, logic helps link these separate pieces of information together into a coherent whole.

Logical Equivalence

The statements

6 is greater than 2 and 2 is less than 6

are two different ways of saying the same thing. Why? Because of the definition of the phrases *greater than* and *less than*. By contrast, although the statements

(1) Dogs bark and cats meow and (2) Cats meow and dogs bark

are also two different ways of saying the same thing, the reason has nothing to do with the definition of the words. It has to do with the logical form of the statements. Any two statements whose logical forms are related in the same way as (1) and (2) would either both be true or both be false. You can see this by examining the following truth table, where the statement variables p and q are substituted for the component statements “Dogs bark” and “Cats meow,” respectively. The table shows that for each combination of truth values for p and q , $p \wedge q$ is true when, and only when, $q \wedge p$ is true. In such a case, the statement forms are called *logically equivalent*, and we say that (1) and (2) are *logically equivalent statements*.

p	q	$p \wedge q$	$q \wedge p$
T	T	T	T
T	F	F	F
F	T	F	F
F	F	F	F

↑ ↑
 $p \wedge q$ and $q \wedge p$ always
 have the same truth
 values, so they are
 logically equivalent

• Definition

Two *statement forms* are called **logically equivalent** if, and only if, they have identical truth values for each possible substitution of statements for their statement variables. The logical equivalence of statement forms P and Q is denoted by writing $P \equiv Q$.

Two *statements* are called **logically equivalent** if, and only if, they have logically equivalent forms when identical component statement variables are used to replace identical component statements.

Testing Whether Two Statement Forms P and Q Are Logically Equivalent

- Construct a truth table with one column for the truth values of P and another column for the truth values of Q .
- Check each combination of truth values of the statement variables to see whether the truth value of P is the same as the truth value of Q .
 - If in each row the truth value of P is the same as the truth value of Q , then P and Q are logically equivalent.
 - If in some row P has a different truth value from Q , then P and Q are not logically equivalent.

Example 2.1.6 Double Negative Property: $\sim(\sim p) \equiv p$

Construct a truth table to show that the negation of the negation of a statement is logically equivalent to the statement, annotating the table with a sentence of explanation.

Solution

p	$\sim p$	$\sim(\sim p)$
T	F	T
F	T	F

p and $\sim(\sim p)$ always have the same truth values, so they are logically equivalent

There are two ways to show that statement forms P and Q are *not* logically equivalent. As indicated previously, one is to use a truth table to find rows for which their truth values differ. The other way is to find concrete statements for each of the two forms, one of which is true and the other of which is false. The next example illustrates both of these ways.

Example 2.1.7 Showing Nonequivalence

Show that the statement forms $\sim(p \wedge q)$ and $\sim p \wedge \sim q$ are not logically equivalent.

Solution

- a. This method uses a truth table annotated with a sentence of explanation.

p	q	$\sim p$	$\sim q$	$p \wedge q$	$\sim(p \wedge q)$	$\sim p \wedge \sim q$
T	T	F	F	T	F	F
T	F	F	T	F	T	F
F	T	T	F	F	T	F
F	F	T	T	F	T	T

$\sim(p \wedge q)$ and $\sim p \wedge \sim q$ have different truth values in rows 2 and 3, so they are not logically equivalent

- b. This method uses an example to show that $\sim(p \wedge q)$ and $\sim p \wedge \sim q$ are not logically equivalent. Let p be the statement “ $0 < 1$ ” and let q be the statement “ $1 < 0$.” Then

$\sim(p \wedge q)$ is “It is not the case that both $0 < 1$ and $1 < 0$,”

which is true. On the other hand,

$$\sim p \wedge \sim q \quad \text{is} \quad "0 \not\leq 1 \quad \text{and} \quad 1 \not\leq 0,"$$

which is false. This example shows that there are concrete statements you can substitute for p and q to make one of the statement forms true and the other false. Therefore, the statement forms are not logically equivalent. ■

Example 2.1.8 Negations of *And* and *Or*: De Morgan's Laws

For the statement “John is tall and Jim is redheaded” to be true, both components must be true. So for the statement to be false, one or both components must be false. Thus the negation can be written as “John is not tall or Jim is not redheaded.” In general, the negation of the conjunction of two statements is logically equivalent to the disjunction of their negations. That is, statements of the forms $\sim(p \wedge q)$ and $\sim p \vee \sim q$ are logically equivalent. Check this using truth tables.

Solution

p	q	$\sim p$	$\sim q$	$p \wedge q$	$\sim(p \wedge q)$	$\sim p \vee \sim q$
T	T	F	F	T	F	F
T	F	F	T	F	T	T
F	T	T	F	F	T	T
F	F	T	T	F	T	T

$\uparrow$ $\uparrow$
 $\sim(p \wedge q)$ and $\sim p \vee \sim q$ always
 have the same truth values, so they
 are logically equivalent



Augustus De Morgan
(1806–1871)

Symbolically,

$$\sim(p \wedge q) \equiv \sim p \vee \sim q.$$

In the exercises at the end of this section you are asked to show the analogous law that the negation of the disjunction of two statements is logically equivalent to the conjunction of their negations:

$$\sim(p \vee q) \equiv \sim p \wedge \sim q.$$

The two logical equivalences of Example 2.1.8 are known as **De Morgan's laws** of logic in honor of Augustus De Morgan, who was the first to state them in formal mathematical terms.

De Morgan's Laws

The negation of an *and* statement is logically equivalent to the *or* statement in which each component is negated.

The negation of an *or* statement is logically equivalent to the *and* statement in which each component is negated.

Example 2.1.9 Applying De Morgan's Laws

Write negations for each of the following statements:

- John is 6 feet tall and he weighs at least 200 pounds.
- The bus was late or Tom's watch was slow.

Solution

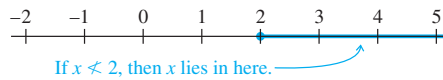
- a. John is not 6 feet tall or he weighs less than 200 pounds.
 b. The bus was not late and Tom's watch was not slow.

Since the statement “neither p nor q ” means the same as “ $\sim p$ and $\sim q$,” an alternative answer for (b) is “Neither was the bus late nor was Tom's watch slow.” ■

If x is a particular real number, saying that x is not less than 2 ($x \not< 2$) means that x does not lie to the left of 2 on the number line. This is equivalent to saying that either $x = 2$ or x lies to the right of 2 on the number line ($x = 2$ or $x > 2$). Hence,

$$x \not< 2 \text{ is equivalent to } x \geq 2.$$

Pictorially,



Similarly,

$$\begin{aligned} x \not> 2 & \text{ is equivalent to } x \leq 2, \\ x \not\leq 2 & \text{ is equivalent to } x > 2, \text{ and} \\ x \not\geq 2 & \text{ is equivalent to } x < 2. \end{aligned}$$

Example 2.1.10 Inequalities and De Morgan's Laws

Use De Morgan's laws to write the negation of $-1 < x \leq 4$.

Solution The given statement is equivalent to

$$-1 < x \text{ and } x \leq 4.$$

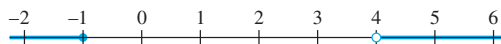
By De Morgan's laws, the negation is

$$-1 \not< x \text{ or } x \not\leq 4,$$

which is equivalent to

$$-1 \geq x \text{ or } x > 4.$$

Pictorially, if $-1 \geq x$ or $x > 4$, then x lies in the shaded region of the number line, as shown below.



Caution! The negation of $-1 < x \leq 4$ is *not* $-1 \not< x \not\leq 4$. It is also not $-1 \geq x > 4$.

De Morgan's laws are frequently used in writing computer programs. For instance, suppose you want your program to delete all files modified outside a certain range of dates, say from date 1 through date 2 inclusive. You would use the fact that

$$\sim(\text{date1} \leq \text{file_modification_date} \leq \text{date2})$$

is equivalent to

$$(\text{file_modification_date} < \text{date1}) \quad \text{or} \quad (\text{date2} < \text{file_modification_date}).$$

Example 2.1.11 A Cautionary Example

According to De Morgan's laws, the negation of

p : Jim is tall and Jim is thin

is

$\sim p$: Jim is not tall or Jim is not thin

because the negation of an *and* statement is the *or* statement in which the two components are negated.

Unfortunately, a potentially confusing aspect of the English language can arise when you are taking negations of this kind. Note that statement p can be written more compactly as

p' : Jim is tall and thin.

When it is so written, another way to negate it is

$\sim(p')$: Jim is not tall and thin.

But in this form the negation looks like an *and* statement. Doesn't that violate De Morgan's laws?

Actually no violation occurs. The reason is that in formal logic the words *and* and *or* are allowed only between complete statements, not between sentence fragments.

One lesson to be learned from this example is that when you apply De Morgan's laws, you must have complete statements on either side of each *and* and on either side of each *or*.



Caution! Although the laws of logic are extremely useful, they should be used as an *aid* to thinking, not as a mechanical substitute for it.

Tautologies and Contradictions

It has been said that all of mathematics reduces to tautologies. Although this is formally true, most working mathematicians think of their subject as having substance as well as form. Nonetheless, an intuitive grasp of basic logical tautologies is part of the equipment of anyone who reasons with mathematics.

• Definition

A **tautology** is a statement form that is always true regardless of the truth values of the individual statements substituted for its statement variables. A statement whose form is a tautology is a **tautological statement**.

A **contradiction** is a statement form that is always false regardless of the truth values of the individual statements substituted for its statement variables. A statement whose form is a contradiction is a **contradictory statement**.

According to this definition, the truth of a tautological statement and the falsity of a contradictory statement are due to the logical structure of the statements themselves and are independent of the meanings of the statements.

Example 2.1.12 Tautologies and Contradictions

Show that the statement form $p \vee \sim p$ is a tautology and that the statement form $p \wedge \sim p$ is a contradiction.

Solution

p	$\sim p$	$p \vee \sim p$	$p \wedge \sim p$
T	F	T	F
F	T	T	F

$\uparrow$ $\uparrow$
 all T's so all F's so
 $p \vee \sim p$ is $p \wedge \sim p$ is a
 a tautology contradiction

Example 2.1.13 Logical Equivalence Involving Tautologies and Contradictions

If **t** is a tautology and **c** is a contradiction, show that $p \wedge \mathbf{t} \equiv p$ and $p \wedge \mathbf{c} \equiv \mathbf{c}$.

Solution

p	t	$p \wedge \mathbf{t}$	p	c	$p \wedge \mathbf{c}$
T	T	T	T	F	F
F	T	F	F	F	F

$\uparrow$ $\uparrow$
 same truth same truth
 values, so values, so
 $p \wedge \mathbf{t} \equiv p$ $p \wedge \mathbf{c} \equiv \mathbf{c}$

Summary of Logical Equivalences

Knowledge of logically equivalent statements is very useful for constructing arguments. It often happens that it is difficult to see how a conclusion follows from one form of a statement, whereas it is easy to see how it follows from a logically equivalent form of the statement. A number of logical equivalences are summarized in Theorem 2.1.1 for future reference.

Theorem 2.1.1 Logical Equivalences

Given any statement variables p, q , and r , a tautology **t** and a contradiction **c**, the following logical equivalences hold.

- | | | |
|----------------------------------|---|---|
| 1. <i>Commutative laws:</i> | $p \wedge q \equiv q \wedge p$ | $p \vee q \equiv q \vee p$ |
| 2. <i>Associative laws:</i> | $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$ | $(p \vee q) \vee r \equiv p \vee (q \vee r)$ |
| 3. <i>Distributive laws:</i> | $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ | $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ |
| 4. <i>Identity laws:</i> | $p \wedge \mathbf{t} \equiv p$ | $p \vee \mathbf{c} \equiv p$ |
| 5. <i>Negation laws:</i> | $p \vee \sim p \equiv \mathbf{t}$ | $p \wedge \sim p \equiv \mathbf{c}$ |
| 6. <i>Double negative law:</i> | $\sim(\sim p) \equiv p$ | |
| 7. <i>Idempotent laws:</i> | $p \wedge p \equiv p$ | $p \vee p \equiv p$ |
| 8. <i>Universal bound laws:</i> | $p \vee \mathbf{t} \equiv \mathbf{t}$ | $p \wedge \mathbf{c} \equiv \mathbf{c}$ |
| 9. <i>De Morgan's laws:</i> | $\sim(p \wedge q) \equiv \sim p \vee \sim q$ | $\sim(p \vee q) \equiv \sim p \wedge \sim q$ |
| 10. <i>Absorption laws:</i> | $p \vee (p \wedge q) \equiv p$ | $p \wedge (p \vee q) \equiv p$ |
| 11. <i>Negations of t and c:</i> | $\sim \mathbf{t} \equiv \mathbf{c}$ | $\sim \mathbf{c} \equiv \mathbf{t}$ |

The proofs of laws 4 and 6, the first parts of laws 1 and 5, and the second part of law 9 have already been given as examples in the text. Proofs of the other parts of the theorem are left as exercises. In fact, it can be shown that the first five laws of Theorem 2.1.1 form a core from which the other laws can be derived. The first five laws are the axioms for a mathematical structure known as a Boolean algebra, which is discussed in Section 6.4.

The equivalences of Theorem 2.1.1 are general laws of thought that occur in all areas of human endeavor. They can also be used in a formal way to rewrite complicated statement forms more simply.

Example 2.1.14 Simplifying Statement Forms

Use Theorem 2.1.1 to verify the logical equivalence

$$\sim(\sim p \wedge q) \wedge (p \vee q) \equiv p.$$

Solution Use the laws of Theorem 2.1.1 to replace sections of the statement form on the left by logically equivalent expressions. Each time you do this, you obtain a logically equivalent statement form. Continue making replacements until you obtain the statement form on the right.

$$\begin{aligned} \sim(\sim p \wedge q) \wedge (p \vee q) &\equiv (\sim(\sim p) \vee \sim q) \wedge (p \vee q) && \text{by De Morgan's laws} \\ &\equiv (p \vee \sim q) \wedge (p \vee q) && \text{by the double negative law} \\ &\equiv p \vee (\sim q \wedge q) && \text{by the distributive law} \\ &\equiv p \vee (q \wedge \sim q) && \text{by the commutative law for } \wedge \\ &\equiv p \vee \mathbf{c} && \text{by the negation law} \\ &\equiv p && \text{by the identity law.} \quad \blacksquare \end{aligned}$$

Skill in simplifying statement forms is useful in constructing logically efficient computer programs and in designing digital logic circuits.

Although the properties in Theorem 2.1.1 can be used to prove the logical equivalence of two statement forms, they cannot be used to prove that statement forms are not logically equivalent. On the other hand, truth tables can always be used to determine both equivalence and nonequivalence, and truth tables are easy to program on a computer. When truth tables are used, however, checking for equivalence always requires 2^n steps, where n is the number of variables. Sometimes you can quickly see that two statement forms are equivalent by Theorem 2.1.1, whereas it would take quite a bit of calculating to show their equivalence using truth tables. For instance, it follows immediately from the associative law for $\wedge$ that $p \wedge (\sim q \wedge \sim r) \equiv (p \wedge \sim q) \wedge \sim r$, whereas a truth table verification requires constructing a table with eight rows.

Test Yourself

Answers to Test Yourself questions are located at the end of each section.

1. An *and* statement is true if, and only if, both components are ____.
2. An *or* statement is false if, and only if, both components are ____.
3. Two statement forms are logically equivalent if, and only if, they always have ____.
4. De Morgan's laws say (1) that the negation of an *and* statement is logically equivalent to the ____ statement in which each component is ____, and (2) that the negation of an *or* statement is logically equivalent to the ____ statement in which each component is ____.
5. A tautology is a statement that is always ____.
6. A contradiction is a statement that is always ____.

Exercise Set 2.1*

In each of 1–4 represent the common form of each argument using letters to stand for component sentences, and fill in the blanks so that the argument in part (b) has the same logical form as the argument in part (a).

- If all integers are rational, then the number 1 is rational.
All integers are rational.
Therefore, the number 1 is rational.
 - If all algebraic expressions can be written in prefix notation, then _____.
_____.
Therefore, $(a + 2b)(a^2 - b)$ can be written in prefix notation.
- If all computer programs contain errors, then this program contains an error.
This program does not contain an error.
Therefore, it is not the case that all computer programs contain errors.
 - If _____, then _____.
2 is not odd.
Therefore, it is not the case that all prime numbers are odd.
- This number is even or this number is odd.
This number is not even.
Therefore, this number is odd.
 - _____ or logic is confusing.
My mind is not shot.
Therefore, _____.
- If n is divisible by 6, then n is divisible by 3.
If n is divisible by 3, then the sum of the digits of n is divisible by 3.
Therefore, if n is divisible by 6, then the sum of the digits of n is divisible by 3.
(Assume that n is a particular, fixed integer.)
 - If this function is _____ then this function is differentiable.
If this function is _____ then this function is continuous.
Therefore, if this function is a polynomial, then this function _____.
- Indicate which of the following sentences are statements.
 - 1,024 is the smallest four-digit number that is a perfect square.
 - She is a mathematics major.
 - $128 = 2^6$
 - $x = 2^6$

Write the statements in 6–9 in symbolic form using the symbols $\sim$, $\vee$, and $\wedge$ and the indicated letters to represent component statements.

- Let s = “stocks are increasing” and i = “interest rates are steady.”

- Stocks are increasing but interest rates are steady.
 - Neither are stocks increasing nor are interest rates steady.
- Juan is a math major but not a computer science major.
(m = “Juan is a math major,” c = “Juan is a computer science major”)
 - Let h = “John is healthy,” w = “John is wealthy,” and s = “John is wise.”
 - John is healthy and wealthy but not wise.
 - John is not wealthy but he is healthy and wise.
 - John is neither healthy, wealthy, nor wise.
 - John is neither wealthy nor wise, but he is healthy.
 - John is wealthy, but he is not both healthy and wise.
 - Either this polynomial has degree 2 or it has degree 3 but not both. (n = “This polynomial has degree 2,” k = “This polynomial has degree 3”)
 - Let p be the statement “DATAENDFLAG is off,” q the statement “ERROR equals 0,” and r the statement “SUM is less than 1,000.” Express the following sentences in symbolic notation.
 - DATAENDFLAG is off, ERROR equals 0, and SUM is less than 1,000.
 - DATAENDFLAG is off but ERROR is not equal to 0.
 - DATAENDFLAG is off; however, ERROR is not 0 or SUM is greater than or equal to 1,000.
 - DATAENDFLAG is on and ERROR equals 0 but SUM is greater than or equal to 1,000.
 - Either DATAENDFLAG is on or it is the case that both ERROR equals 0 and SUM is less than 1,000.

- In the following sentence, is the word *or* used in its inclusive or exclusive sense? A team wins the playoffs if it wins two games in a row or a total of *three* games.

Write truth tables for the statement forms in 12–15.

- $\sim p \wedge q$
- $\sim(p \wedge q) \vee (p \vee q)$
- $p \wedge (q \wedge r)$
- $p \wedge (\sim q \vee r)$

Determine whether the statement forms in 16–24 are logically equivalent. In each case, construct a truth table and include a sentence justifying your answer. Your sentence should show that you understand the meaning of logical equivalence.

- $p \vee (p \wedge q)$ and p
- $\sim(p \wedge q)$ and $\sim p \wedge \sim q$
- $p \vee \mathbf{t}$ and $\mathbf{t}$
- $p \wedge \mathbf{t}$ and p
- $p \wedge \mathbf{c}$ and $p \vee \mathbf{c}$
- $(p \wedge q) \wedge r$ and $p \wedge (q \wedge r)$

*For exercises with blue numbers or letters, solutions are given in Appendix B. The symbol **H** indicates that only a hint or a partial solution is given. The symbol ***** signals that an exercise is more challenging than usual.

22. $p \wedge (q \vee r)$ and $(p \wedge q) \vee (p \wedge r)$

23. $(p \wedge q) \vee r$ and $p \wedge (q \vee r)$

24. $(p \vee q) \vee (p \wedge r)$ and $(p \vee q) \wedge r$

Use De Morgan's laws to write negations for the statements in 25–31.

25. Hal is a math major and Hal's sister is a computer science major.

26. Sam is an orange belt and Kate is a red belt.

27. The connector is loose or the machine is unplugged.

28. The units digit of 4^{67} is 4 or it is 6.

29. This computer program has a logical error in the first ten lines or it is being run with an incomplete data set.

30. The dollar is at an all-time high and the stock market is at a record low.

31. The train is late or my watch is fast.

Assume x is a particular real number and use De Morgan's laws to write negations for the statements in 32–37.

32. $-2 < x < 7$

33. $-10 < x < 2$

34. $x < 2$ or $x > 5$

35. $x \leq -1$ or $x > 1$

36. $1 > x \geq -3$

37. $0 > x \geq -7$

In 38 and 39, imagine that num_orders and $num_instock$ are particular values, such as might occur during execution of a computer program. Write negations for the following statements.

38. $(num_orders > 100 \text{ and } num_instock \leq 500) \text{ or } num_instock < 200$

39. $(num_orders < 50 \text{ and } num_instock > 300) \text{ or } (50 \leq num_orders < 75 \text{ and } num_instock > 500)$

Use truth tables to establish which of the statement forms in 40–43 are tautologies and which are contradictions.

40. $(p \wedge q) \vee (\sim p \vee (p \wedge \sim q))$

41. $(p \wedge \sim q) \wedge (\sim p \vee q)$

42. $((\sim p \wedge q) \wedge (q \wedge r)) \wedge \sim q$

43. $(\sim p \vee q) \vee (p \wedge \sim q)$

In 44 and 45, determine whether the statements in (a) and (b) are logically equivalent.

44. Assume x is a particular real number.

a. $x < 2$ or it is not the case that $1 < x < 3$.

b. $x \leq 1$ or either $x < 2$ or $x \geq 3$.

45. a. Bob is a double math and computer science major and Ann is a math major, but Ann is not a double math and computer science major.

b. It is not the case that both Bob and Ann are double math and computer science majors, but it is the case that Ann is a math major and Bob is a double math and computer science major.

★ 46. In Example 2.1.4, the symbol $\oplus$ was introduced to denote *exclusive or*, so $p \oplus q \equiv (p \vee q) \wedge \sim(p \wedge q)$. Hence the truth table for *exclusive or* is as follows:

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

a. Find simpler statement forms that are logically equivalent to $p \oplus p$ and $(p \oplus p) \oplus p$.

b. Is $(p \oplus q) \oplus r \equiv p \oplus (q \oplus r)$? Justify your answer.

c. Is $(p \oplus q) \wedge r \equiv (p \wedge r) \oplus (q \wedge r)$? Justify your answer.

★ 47. In logic and in standard English, a double negative is equivalent to a positive. There is one fairly common English usage in which a “double positive” is equivalent to a negative. What is it? Can you think of others?

In 48 and 49 below, a logical equivalence is derived from Theorem 2.1.1. Supply a reason for each step.

$$\begin{aligned}
 48. \quad (p \wedge \sim q) \vee (p \wedge q) &\equiv p \wedge (\sim q \vee q) && \text{by (a)} \\
 &\equiv p \wedge (q \vee \sim q) && \text{by (b)} \\
 &\equiv p \wedge \mathbf{t} && \text{by (c)} \\
 &\equiv p && \text{by (d)}
 \end{aligned}$$

Therefore, $(p \wedge \sim q) \vee (p \wedge q) \equiv p$.

$$\begin{aligned}
 49. \quad (p \vee \sim q) \wedge (\sim p \vee \sim q) & \\
 &\equiv (\sim q \vee p) \wedge (\sim q \vee \sim p) && \text{by (a)} \\
 &\equiv \sim q \vee (p \wedge \sim p) && \text{by (b)} \\
 &\equiv \sim q \vee \mathbf{f} && \text{by (c)} \\
 &\equiv \sim q && \text{by (d)}
 \end{aligned}$$

Therefore, $(p \vee \sim q) \wedge (\sim p \vee \sim q) \equiv \sim q$.

Use Theorem 2.1.1 to verify the logical equivalences in 50–54. Supply a reason for each step.

50. $(p \wedge \sim q) \vee p \equiv p$ 51. $p \wedge (\sim q \vee p) \equiv p$

52. $\sim(p \vee \sim q) \vee (\sim p \wedge \sim q) \equiv \sim p$

53. $\sim((\sim p \wedge q) \vee (\sim p \wedge \sim q)) \vee (p \wedge q) \equiv p$

54. $(p \wedge (\sim(\sim p \vee q))) \vee (p \wedge q) \equiv p$

Answers for Test Yourself

1. true 2. false 3. the same truth values 4. *or*; negated; *and*; negated 5. true 6. false

2.2 Conditional Statements

... hypothetical reasoning implies the subordination of the real to the realm of the possible ... — Jean Piaget, 1972

When you make a logical inference or deduction, you reason *from* a hypothesis *to* a conclusion. Your aim is to be able to say, “If such and such is known, *then* something or other must be the case.”

Let p and q be statements. A sentence of the form “If p then q ” is denoted symbolically by “ $p \rightarrow q$ ”; p is called the *hypothesis* and q is called the *conclusion*. For instance, consider the following statement:

If 4,686 is divisible by 6, then 4,686 is divisible by 3

hypothesis
conclusion

Such a sentence is called *conditional* because the truth of statement q is conditioned on the truth of statement p .

The notation $p \rightarrow q$ indicates that $\rightarrow$ is a connective, like $\wedge$ or $\vee$, that can be used to join statements to create new statements. To define $p \rightarrow q$ as a statement, therefore, we must specify the truth values for $p \rightarrow q$ as we specified truth values for $p \wedge q$ and for $p \vee q$. As is the case with the other connectives, the formal definition of truth values for $\rightarrow$ (if-then) is based on its everyday, intuitive meaning. Consider an example.

Suppose you go to interview for a job at a store and the owner of the store makes you the following promise:

If you show up for work Monday morning, then you will get the job.

Under what circumstances are you justified in saying the owner spoke falsely? That is, under what circumstances is the above sentence false? The answer is: You *do* show up for work Monday morning and you do *not* get the job.

After all, the owner’s promise only says you will get the job *if* a certain condition (showing up for work Monday morning) is met; it says nothing about what will happen if the condition is *not* met. So if the condition is not met, you cannot in fairness say the promise is false regardless of whether or not you get the job.

The above example was intended to convince you that *the only combination of circumstances in which you would call a conditional sentence false occurs when the hypothesis is true and the conclusion is false*. In all other cases, you would not call the sentence false. This implies that the only row of the truth table for $p \rightarrow q$ that should be filled in with an F is the row where p is T and q is F. No other row should contain an F. But each row of a truth table must be filled in with either a T or an F. Thus all other rows of the truth table for $p \rightarrow q$ must be filled in with T’s.

Truth Table for $p \rightarrow q$

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

• Definition

If p and q are statement variables, the **conditional** of q by p is “If p then q ” or “ p implies q ” and is denoted $p \rightarrow q$. It is false when p is true and q is false; otherwise it is true. We call p the **hypothesis** (or **antecedent**) of the conditional and q the **conclusion** (or **consequent**).

A conditional statement that is true by virtue of the fact that its hypothesis is false is often called **vacuously true** or **true by default**. Thus the statement “If you show up for work Monday morning, then you will get the job” is vacuously true if you do not show up for work Monday morning. In general, when the “if” part of an if-then statement is false, the statement as a whole is said to be true, regardless of whether the conclusion is true or false.

Example 2.2.1 A Conditional Statement with a False Hypothesis

Consider the statement:

If $0 = 1$ then $1 = 2$.

As strange as it may seem, since the hypothesis of this statement is false, the statement as a whole is true. ■

The philosopher Willard Van Orman Quine advises against using the phrase “ p implies q ” to mean “ $p \rightarrow q$ ” because the word *implies* suggests that q can be logically deduced from p and this is often not the case. Nonetheless, the phrase is used by many people, probably because it is a convenient replacement for the $\rightarrow$ symbol. And, of course, in many cases a conclusion can be deduced from a hypothesis, even when the hypothesis is false.

In expressions that include $\rightarrow$ as well as other logical operators such as $\wedge$, $\vee$, and $\sim$, the **order of operations** is that $\rightarrow$ is performed last. Thus, according to the specification of order of operations in Section 2.1, $\sim$ is performed first, then $\wedge$ and $\vee$, and finally $\rightarrow$.

Note For example, if $0 = 1$, then, by adding 1 to both sides of the equation, you can deduce that $1 = 2$.

Example 2.2.2 Truth Table for $p \vee \sim q \rightarrow \sim p$

Construct a truth table for the statement form $p \vee \sim q \rightarrow \sim p$.

Solution By the order of operations given above, the following two expressions are equivalent: $p \vee \sim q \rightarrow \sim p$ and $(p \vee (\sim q)) \rightarrow (\sim p)$, and this order governs the construction of the truth table. First fill in the four possible combinations of truth values for p and q , and then enter the truth values for $\sim p$ and $\sim q$ using the definition of negation. Next fill in the $p \vee \sim q$ column using the definition of $\vee$. Finally, fill in the $p \vee \sim q \rightarrow \sim p$ column using the definition of $\rightarrow$. The only rows in which the hypothesis $p \vee \sim q$ is true and the conclusion $\sim p$ is false are the first and second rows. So you put F’s in those two rows and T’s in the other two rows.

		conclusion		hypothesis	
p	q	$\sim p$	$\sim q$	$p \vee \sim q$	$p \vee \sim q \rightarrow \sim p$
T	T	F	F	T	F
T	F	F	T	T	F
F	T	T	F	F	T
F	F	T	T	T	T

Example 2.2.3 Division into Cases: Showing that $p \vee q \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r)$

Solution First fill in the eight possible combinations of truth values for p , q , and r . Then fill in the columns for $p \vee q$, $p \rightarrow r$, and $q \rightarrow r$ using the definitions of *or* and *if-then*. For instance, the $p \rightarrow r$ column has F's in the second and fourth rows because these are the rows in which p is true and q is false. Next fill in the $p \vee q \rightarrow r$ column using the definition of *if-then*. The rows in which the hypothesis $p \vee q$ is true and the conclusion r is false are the second, fourth, and sixth. So F's go in these rows and T's in all the others. The complete table shows that $p \vee q \rightarrow r$ and $(p \rightarrow r) \wedge (q \rightarrow r)$ have the same truth values for each combination of truth values of p , q , and r . Hence the two statement forms are logically equivalent.

p	q	r	$p \vee q$	$p \rightarrow r$	$q \rightarrow r$	$p \vee q \rightarrow r$	$(p \rightarrow r) \wedge (q \rightarrow r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	F
T	F	T	T	T	T	T	T
T	F	F	T	F	T	F	F
F	T	T	T	T	T	T	T
F	T	F	T	T	F	F	F
F	F	T	F	T	T	T	T
F	F	F	F	T	T	T	T

Representation of If-Then As Or

$$p \rightarrow q \equiv \sim p \vee q.$$

Example 2.2.4 Application of the Equivalence between $\sim p \vee q$ and $p \rightarrow q$

Either you get to work on time or you are fired.

You get to work on time.

and q be

You are fired.

Then the given statement is $\sim p \vee q$. Also p is

You do not get to work on time.

So the equivalent if-then version, $p \rightarrow q$, is

If you do not get to work on time, then you are fired. ■

The Negation of a Conditional Statement

By definition, $p \rightarrow q$ is false if, and only if, its hypothesis, p , is true and its conclusion, q , is false. It follows that

The negation of “if p then q ” is logically equivalent to “ p and not q .”

This can be restated symbolically as follows:

$$\sim(p \rightarrow q) \equiv p \wedge \sim q$$

You can also obtain this result by starting from the logical equivalence $p \rightarrow q \equiv \sim p \vee q$. Take the negation of both sides to obtain

$$\begin{aligned} \sim(p \rightarrow q) &\equiv \sim(\sim p \vee q) \\ &\equiv \sim(\sim p) \wedge (\sim q) && \text{by De Morgan's laws} \\ &\equiv p \wedge \sim q && \text{by the double negative law.} \end{aligned}$$

Yet another way to derive this result is to construct truth tables for $\sim(p \rightarrow q)$ and for $p \wedge \sim q$ and to check that they have the same truth values. (See exercise 13(b) at the end of this section.)

Example 2.2.5 Negations of If-Then Statements

Write negations for each of the following statements:

- If my car is in the repair shop, then I cannot get to class.
- If Sara lives in Athens, then she lives in Greece.

Solution

- My car is in the repair shop and I can get to class.
- Sara lives in Athens and she does not live in Greece. (Sara might live in Athens, Georgia; Athens, Ohio; or Athens, Wisconsin.) ■

It is tempting to write the negation of an if-then statement as another if-then statement. Please resist that temptation!



Caution! Remember that the negation of an if-then statement does not start with the word *if*.

The Contrapositive of a Conditional Statement

One of the most fundamental laws of logic is the equivalence between a conditional statement and its contrapositive.

• Definition

The **contrapositive** of a conditional statement of the form “If p then q ” is

If $\sim q$ then $\sim p$.

Symbolically,

The contrapositive of $p \rightarrow q$ is $\sim q \rightarrow \sim p$.

The fact is that

A conditional statement is logically equivalent to its contrapositive.

You are asked to establish this equivalence in exercise 26 at the end of this section.

Example 2.2.6 Writing the Contrapositive

Write each of the following statements in its equivalent contrapositive form:

- a. If Howard can swim across the lake, then Howard can swim to the island.
- b. If today is Easter, then tomorrow is Monday.

Solution

- a. If Howard cannot swim to the island, then Howard cannot swim across the lake.
- b. If tomorrow is not Monday, then today is not Easter. ■

When you are trying to solve certain problems, you may find that the contrapositive form of a conditional statement is easier to work with than the original statement. Replacing a statement by its contrapositive may give the extra push that helps you over the top in your search for a solution. This logical equivalence is also the basis for one of the most important laws of deduction, modus tollens (to be explained in Section 2.3), and for the contrapositive method of proof (to be explained in Section 4.6).

The Converse and Inverse of a Conditional Statement

The fact that a conditional statement and its contrapositive are logically equivalent is very important and has wide application. Two other variants of a conditional statement are *not* logically equivalent to the statement.

• Definition

Suppose a conditional statement of the form “If p then q ” is given.

1. The **converse** is “If q then p .”
2. The **inverse** is “If $\sim p$ then $\sim q$.”

Symbolically,

The converse of $p \rightarrow q$ is $q \rightarrow p$,

and

The inverse of $p \rightarrow q$ is $\sim p \rightarrow \sim q$.

Example 2.2.7 Writing the Converse and the Inverse

Write the converse and inverse of each of the following statements:

- a. If Howard can swim across the lake, then Howard can swim to the island.
- b. If today is Easter, then tomorrow is Monday.

Solution

- a. *Converse:* If Howard can swim to the island, then Howard can swim across the lake.
Inverse: If Howard cannot swim across the lake, then Howard cannot swim to the island.
- b. *Converse:* If tomorrow is Monday, then today is Easter.
Inverse: If today is not Easter, then tomorrow is not Monday. ■



Caution! Many people believe that if a conditional statement is true, then its converse and inverse must also be true. This is not correct!

Note that while the statement “If today is Easter, then tomorrow is Monday” is always true, both its converse and inverse are false on every Sunday except Easter.

1. A conditional statement and its converse are *not* logically equivalent.
2. A conditional statement and its inverse are *not* logically equivalent.
3. The converse and the inverse of a conditional statement are logically equivalent to each other.

In exercises 24, 25, and 27 at the end of this section, you are asked to use truth tables to verify the statements in the box above. Note that the truth of statement 3 also follows from the observation that the inverse of a conditional statement is the contrapositive of its converse.

Only If and the Biconditional

To say “ p only if q ” means that p can take place *only* if q takes place also. That is, if q does not take place, then p cannot take place. Another way to say this is that if p occurs, then q must also occur (by the logical equivalence between a statement and its contrapositive).

• Definition

It p and q are statements,

p **only if** q means “if not q then not p ,”

or, equivalently,

“if p then q .”

Example 2.2.8 Converting Only If to If-Then

Rewrite the following statement in if-then form in two ways, one of which is the contrapositive of the other.

John will break the world’s record for the mile run only if he runs the mile in under four minutes.

Solution *Version 1:* If John does not run the mile in under four minutes, then he will not break the world’s record.

Version 2: If John breaks the world’s record, then he will have run the mile in under four minutes. ■



Caution! “ p only if q ” does *not* mean “ p if q .”

Note that it is possible for “ p only if q ” to be true at the same time that “ p if q ” is false. For instance, to say that John will break the world’s record only if he runs the mile in under four minutes does not mean that John will break the world’s record if he runs the mile in under four minutes. His time could be under four minutes but still not be fast enough to break the record.

• Definition

Given statement variables p and q , the **biconditional of p and q** is “ p if, and only if, q ” and is denoted $p \leftrightarrow q$. It is true if both p and q have the same truth values and is false if p and q have opposite truth values. The words *if and only if* are sometimes abbreviated **iff**.

The biconditional has the following truth table:

Truth Table for $p \leftrightarrow q$

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

In order of operations $\leftrightarrow$ is coequal with $\rightarrow$. As with $\wedge$ and $\vee$, the only way to indicate precedence between them is to use parentheses. The full hierarchy of operations for the five logical operators is on the next page.

The occurrence of r is *necessary* to obtain the occurrence of s . Note that because of the equivalence between a statement and its contrapositive,

r is a necessary condition for s also means “if s then r .”

Consequently,

r is a necessary and sufficient condition for s means “ r if, and only if, s .”

Example 2.2.10 Interpreting Necessary and Sufficient Conditions

Consider the statement “If John is eligible to vote, then he is at least 18 years old.” The truth of the condition “John is eligible to vote” is *sufficient* to ensure the truth of the condition “John is at least 18 years old.” In addition, the condition “John is at least 18 years old” is *necessary* for the condition “John is eligible to vote” to be true. If John were younger than 18, then he would not be eligible to vote. ■

Example 2.2.11 Converting a Sufficient Condition to If-Then Form

Rewrite the following statement in the form “If A then B ”:

Pia’s birth on U.S. soil is a sufficient condition
for her to be a U.S. citizen.

Solution If Pia was born on U.S. soil, then she is a U.S. citizen. ■

Example 2.2.12 Converting a Necessary Condition to If-Then Form

Use the contrapositive to rewrite the following statement in two ways:

George’s attaining age 35 is a necessary condition
for his being president of the United States.

Solution *Version 1:* If George has not attained the age of 35, then he cannot be president of the United States.

Version 2: If George can be president of the United States, then he has attained the age of 35. ■

Remarks

1. *In logic, a hypothesis and conclusion are not required to have related subject matters.*
In ordinary speech we never say things like “If computers are machines, then Babe Ruth was a baseball player” or “If $2 + 2 = 5$, then Mickey Mouse is president of the United States.” We formulate a sentence like “If p then q ” only if there is some connection of content between p and q .
In logic, however, the two parts of a conditional statement need not have related meanings. The reason? If there were such a requirement, who would enforce it? What one person perceives as two unrelated clauses may seem related to someone else. There would have to be a central arbiter to check each conditional sentence before anyone could use it, to be sure its clauses were in proper relation. This is impractical, to say the least!

Thus a statement like “if computers are machines, then Babe Ruth was a baseball player” is allowed, and it is even called true because both its hypothesis and its conclusion are true. Similarly, the statement “If $2 + 2 = 5$, then Mickey Mouse is president of the United States” is allowed and is called true because its hypothesis is false, even though doing so may seem ridiculous.

In mathematics it often happens that a carefully formulated definition that successfully covers the situations for which it was primarily intended is later seen to be satisfied by some extreme cases that the formulator did not have in mind. But those are the breaks, and it is important to get into the habit of exploring definitions fully to seek out and understand *all* their instances, even the unusual ones.

2. *In informal language, simple conditionals are often used to mean biconditionals.*

The formal statement “ p if, and only if, q ” is seldom used in ordinary language. Frequently, when people intend the biconditional they leave out either the *and only if* or the *if and*. That is, they say either “ p if q ” or “ p only if q ” when they really mean “ p if, and only if, q .” For example, consider the statement “You will get dessert if, and only if, you eat your dinner.” Logically, this is equivalent to the conjunction of the following two statements.

Statement 1: If you eat your dinner, then you will get dessert.

Statement 2: You will get dessert only if you eat your dinner.

or

If you do not eat your dinner, then you will not get dessert.

Now how many parents in the history of the world have said to their children “You will get dessert if, and only if, you eat your dinner”? Not many! Most say either “If you eat your dinner, you will get dessert” (these take the positive approach—they emphasize the reward) or “You will get dessert only if you eat your dinner” (these take the negative approach—they emphasize the punishment). Yet the parents who promise the reward intend to suggest the punishment as well, and those who threaten the punishment will certainly give the reward if it is earned. Both sets of parents expect that their conditional statements will be interpreted as biconditionals.

Since we often (correctly) interpret conditional statements as biconditionals, it is not surprising that we may come to believe (mistakenly) that conditional statements are always logically equivalent to their inverses and converses. In formal settings, however, statements must have unambiguous interpretations. If-then statements can’t sometimes mean “if-then” and other times mean “if and only if.” When using language in mathematics, science, or other situations where precision is important, it is essential to interpret if-then statements according to the formal definition and not to confuse them with their converses and inverses.

Test Yourself

1. An *if-then* statement is false if, and only if, the hypothesis is ____ and the conclusion is ____.
2. The negation of “if p then q ” is ____.
3. The converse of “if p then q ” is ____.
4. The contrapositive of “if p then q ” is ____.
5. The inverse of “if p then q ” is ____.
6. A conditional statement and its contrapositive are ____.
7. A conditional statement and its converse are not ____.
8. “ R is a sufficient condition for S ” means “if ____ then ____.”
9. “ R is a necessary condition for S ” means “if ____ then ____.”
10. “ R only if S ” means “if ____ then ____.”

Exercise Set 2.2

Rewrite the statements in 1–4 in if-then form.

1. This loop will repeat exactly N times if it does not contain a **stop** or a **go to**.
2. I am on time for work if I catch the 8:05 bus.
3. Freeze or I'll shoot.
4. Fix my ceiling or I won't pay my rent.

Construct truth tables for the statement forms in 5–11.

5. $\sim p \vee q \rightarrow \sim q$
6. $(p \vee q) \vee (\sim p \wedge q) \rightarrow q$
7. $p \wedge \sim q \rightarrow r$
8. $\sim p \vee q \rightarrow r$
9. $p \wedge \sim r \leftrightarrow q \vee r$
10. $(p \rightarrow r) \leftrightarrow (q \rightarrow r)$

$$11. (p \rightarrow (q \rightarrow r)) \leftrightarrow ((p \wedge q) \rightarrow r)$$

12. Use the logical equivalence established in Example 2.2.3, $p \vee q \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r)$, to rewrite the following statement. (Assume that x represents a fixed real number.)

If $x > 2$ or $x < -2$, then $x^2 > 4$.

13. Use truth tables to verify the following logical equivalences. Include a few words of explanation with your answers.

$$\text{a. } p \rightarrow q \equiv \sim p \vee q \quad \text{b. } \sim(p \rightarrow q) \equiv p \wedge \sim q.$$

- H** 14. **a.** Show that the following statement forms are all logically equivalent.

$$p \rightarrow q \vee r, \quad p \wedge \sim q \rightarrow r, \quad \text{and} \quad p \wedge \sim r \rightarrow q$$

- b.** Use the logical equivalences established in part (a) to rewrite the following sentence in two different ways. (Assume that n represents a fixed integer.)

If n is prime, then n is odd or n is 2.

15. Determine whether the following statement forms are logically equivalent:

$$p \rightarrow (q \rightarrow r) \quad \text{and} \quad (p \rightarrow q) \rightarrow r$$

In 16 and 17, write each of the two statements in symbolic form and determine whether they are logically equivalent. Include a truth table and a few words of explanation.

16. If you paid full price, you didn't buy it at Crown Books. You didn't buy it at Crown Books or you paid full price.
17. If 2 is a factor of n and 3 is a factor of n , then 6 is a factor of n . 2 is not a factor of n or 3 is not a factor of n or 6 is a factor of n .
18. Write each of the following three statements in symbolic form and determine which pairs are logically equivalent. Include truth tables and a few words of explanation.

If it walks like a duck and it talks like a duck, then it is a duck.

Either it does not walk like a duck or it does not talk like a duck, or it is a duck.

If it does not walk like a duck and it does not talk like a duck, then it is not a duck.

19. True or false? The negation of "If Sue is Luiz's mother, then Ali is his cousin" is "If Sue is Luiz's mother, then Ali is not his cousin."
20. Write negations for each of the following statements. (Assume that all variables represent fixed quantities or entities, as appropriate.)
 - a. If P is a square, then P is a rectangle.
 - b. If today is New Year's Eve, then tomorrow is January.
 - c. If the decimal expansion of r is terminating, then r is rational.
 - d. If n is prime, then n is odd or n is 2.
 - e. If x is nonnegative, then x is positive or x is 0.
 - f. If Tom is Ann's father, then Jim is her uncle and Sue is her aunt.
 - g. If n is divisible by 6, then n is divisible by 2 and n is divisible by 3.
21. Suppose that p and q are statements so that $p \rightarrow q$ is false. Find the truth values of each of the following:

$$\text{a. } \sim p \rightarrow q \quad \text{b. } p \vee q \quad \text{c. } q \rightarrow p$$

- H** 22. Write contrapositives for the statements of exercise 20.

- H** 23. Write the converse and inverse for each statement of exercise 20.

Use truth tables to establish the truth of each statement in 24–27.

24. A conditional statement is not logically equivalent to its converse.
25. A conditional statement is not logically equivalent to its inverse.
26. A conditional statement and its contrapositive are logically equivalent to each other.
27. The converse and inverse of a conditional statement are logically equivalent to each other.

- H** 28. "Do you mean that you think you can find out the answer to it?" said the March Hare.

"Exactly so," said Alice.

"Then you should say what you mean," the March Hare went on.

"I do," Alice hastily replied; "at least—at least I mean what I say—that's the same thing, you know."

“Not the same thing a bit!” said the Hatter. “Why, you might just as well say that ‘I see what I eat’ is the same thing as ‘I eat what I see’!”

—from “A Mad Tea-Party” in *Alice in Wonderland*,
by Lewis Carroll

The Hatter is right. “I say what I mean” is not the same thing as “I mean what I say.” Rewrite each of these two sentences in if-then form and explain the logical relation between them. (This exercise is referred to in the introduction to Chapter 4.)

If statement forms P and Q are logically equivalent, then $P \leftrightarrow Q$ is a tautology. Conversely, if $P \leftrightarrow Q$ is a tautology, then P and Q are logically equivalent. Use $\leftrightarrow$ to convert each of the logical equivalences in 29–31 to a tautology. Then use a truth table to verify each tautology.

$$29. p \rightarrow (q \vee r) \equiv (p \wedge \sim q) \rightarrow r$$

$$30. p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

$$31. p \rightarrow (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$$

Rewrite each of the statements in 32 and 33 as a conjunction of two if-then statements.

32. This quadratic equation has two distinct real roots if, and only if, its discriminant is greater than zero.

33. This integer is even if, and only if, it equals twice some integer.

Rewrite the statements in 34 and 35 in if-then form in two ways, one of which is the contrapositive of the other.

34. The Cubs will win the pennant only if they win tomorrow’s game.

35. Sam will be allowed on Signe’s racing boat only if he is an expert sailor.

36. Taking the long view on your education, you go to the Prestige Corporation and ask what you should do in college to be hired when you graduate. The personnel director replies that you will be hired *only if* you major in mathematics or computer science, get a B average or better, and take accounting. You do, in fact, become a math major, get a B⁺ average, and take accounting. You return to Prestige Corporation, make a formal application, and are turned down. Did the personnel director lie to you?

Some programming languages use statements of the form “ r unless s ” to mean that as long as s does not happen, then r will happen. More formally:

Definition: If r and s are statements,
 r unless s means if $\sim s$ then r .

In 37–39, rewrite the statements in if-then form.

37. Payment will be made on the fifth unless a new hearing is granted.

38. Ann will go unless it rains.

39. This door will not open unless a security code is entered.

Rewrite the statements in 40 and 41 in if-then form.

40. Catching the 8:05 bus is a sufficient condition for my being on time for work.

41. Having two 45° angles is a sufficient condition for this triangle to be a right triangle.

Use the contrapositive to rewrite the statements in 42 and 43 in if-then form in two ways.

42. Being divisible by 3 is a necessary condition for this number to be divisible by 9.

43. Doing homework regularly is a necessary condition for Jim to pass the course.

Note that “a sufficient condition for s is r ” means r is a sufficient condition for s and that “a necessary condition for s is r ” means r is a necessary condition for s . Rewrite the statements in 44 and 45 in if-then form.

44. A sufficient condition for Jon’s team to win the championship is that it win the rest of its games.

45. A necessary condition for this computer program to be correct is that it not produce error messages during translation.

46. “If compound X is boiling, then its temperature must be at least 150°C.” Assuming that this statement is true, which of the following must also be true?

- a. If the temperature of compound X is at least 150°C, then compound X is boiling.
- b. If the temperature of compound X is less than 150°C, then compound X is not boiling.
- c. Compound X will boil only if its temperature is at least 150°C.
- d. If compound X is not boiling, then its temperature is less than 150°C.
- e. A necessary condition for compound X to boil is that its temperature be at least 150°C.
- f. A sufficient condition for compound X to boil is that its temperature be at least 150°C.

In 47–50 (a) use the logical equivalences $p \rightarrow q \equiv \sim p \vee q$ and $p \leftrightarrow q \equiv (\sim p \vee q) \wedge (\sim q \vee p)$ to rewrite the given statement forms without using the symbol $\rightarrow$ or $\leftrightarrow$, and (b) use the logical equivalence $p \vee q \equiv \sim(\sim p \wedge \sim q)$ to rewrite each statement form using only $\wedge$ and $\sim$.

$$47. p \wedge \sim q \rightarrow r \qquad 48. p \vee \sim q \rightarrow r \vee q$$

$$49. (p \rightarrow r) \leftrightarrow (q \rightarrow r)$$

$$50. (p \rightarrow (q \rightarrow r)) \leftrightarrow ((p \wedge q) \rightarrow r)$$

51. Given any statement form, is it possible to find a logically equivalent form that uses only $\sim$ and $\wedge$? Justify your answer.